UNIT III – Milling, Grinding, Finishing, Jigs & Fixtures

23A03501 Machining Processes · B.Tech Mech · Handwritten exam notes

1. MILLING MACHINE

Milling = machining with a **multi-point rotating cutter** fed against a work moving (table) – each tooth removes a small chip intermittently.

Principle of working

- Cutter rotates on spindle/arbor (**main motion**)
- Work clamped on table → **feed** (long., cross, vertical)
- Cutting speed $V = \pi DN/1000$ m/min ; feed → per tooth f_t, per rev, per min

Specifications of a milling m/c

- Table size (length × width) & max. travels – long., cross, vertical
- Spindle taper (ISO 40/50), spindle speeds range & no. of feeds

Parts of column & knee type

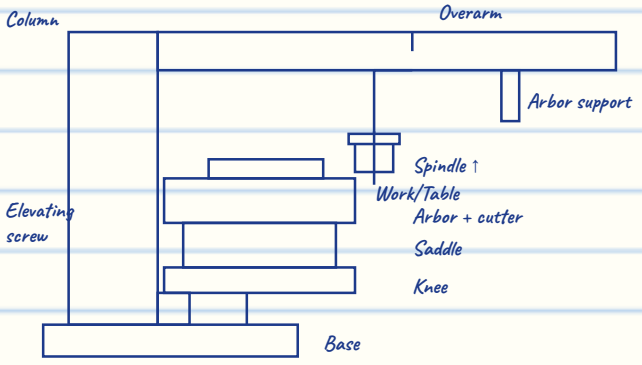


Fig 1: Column & knee milling machine (schematic)

- **Base** – foundation, holds coolant; **Column** – houses spindle drive, speed gear box
- **Knee** – slides on column, vertical motion by elevating screw
- **Saddle** – on knee, cross-feed; **Table** – on saddle, long. feed (T-slots)
- **Arbor** – shaft holding cutter; **Arbor support** – avoids deflection

2. Classification of milling machines

Type	Features / use
Column & knee	Most common, tool-room & general work. (a) **Plain** (horizontal spindle, 3 motions) (b) **Universal** (table swivels ±45° → helical work) (c) **Vertical** (spindle vertical, end mills, T-slots, dies)
Bed type	Table on rigid bed, only long. motion; spindle head moves vertically; heavy, mass production. Simplex / duplex / triplex
Planer type	Like planer; 2–4 heads on cross-rail & columns; heavy big jobs (machine beds)
Special type	Rotary table, drum, profile (copy), pantograph, gear hobbing, thread, tracer, CNC milling

3. Milling methods

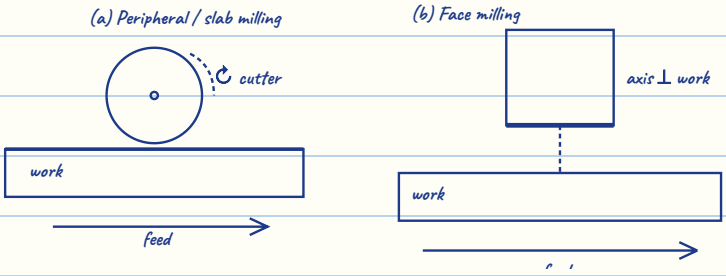


Fig 2: Peripheral and face milling

- **Peripheral (slab) milling** – teeth on periphery; cutter axis ∥ work surface; plain/helical cutter, horizontal mill; good finish for flat/wide surface
- **Face milling** – teeth on face & periphery; axis ⊥ to work; large flat surfaces, high MRR, inserted-tooth cutters
- **End milling** – slots, pockets, profiles (vertical mill)

Up (conventional) vs Down (climb) milling

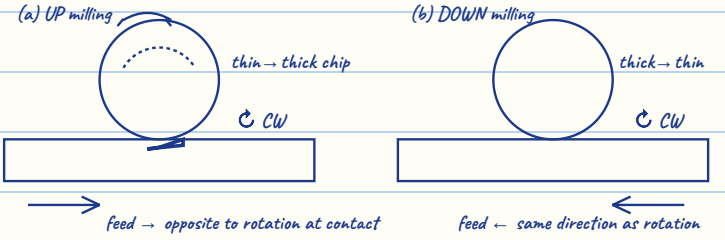


Fig 3: Up vs down milling (cutter ↻, work direction differs)

Point	Up (conventional)	Down (climb)
Feed vs rotation	Opposite at contact	Same direction
Chip thickness	Zero → max (thin start)	Max → zero
Cutting force	Lifts work, tends to loosen	Pushes work down on table
Surface finish	Poorer (rubbing, BUE)	Better
Tool life	Less (rubbing, scale)	More
Backlash	Not harmful	Needs **backlash eliminator**
Use	Rough, castings, forgings (hard skin)	Finishing, thin & rigid, CNC

4. Milling cutters

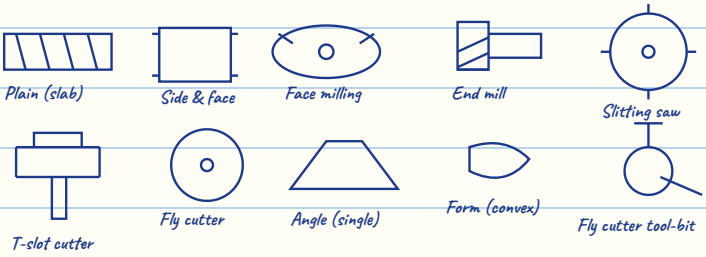


Fig 4: Common milling cutters

- **Plain (slab)** – cylindrical, teeth on periphery; helical teeth → smooth cut
- **Side & face** – teeth on periphery + one/both sides; slots, steps; **staggered** type for deep slots
- **Face mill** – large dia, inserted carbide tips; **End mill** – shank, teeth on end+periphery
- **Slitting saw** – thin (0.5–5 mm), parting & deep narrow slots
- **T-slot cutter** – end mill + small neck & wide head; cuts T-slots after slot
- **Angle** – single/double angle (45°,60°) → dovetails, V-grooves, ratchet
- **Form cutters** – convex, concave, corner-rounding, gear (involute) cutters
- **Fly cutter** – single point bit in arbor; cheap special profile / facing

Cutter nomenclature (plain milling cutter)

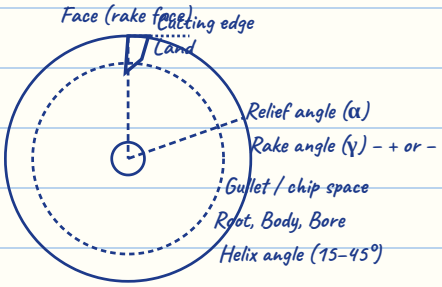


Fig 5: Nomenclature of milling cutter

5. Milling operations

- **Plain/slab milling** – flat surface · **Face milling** – flat ⊥ axis
- **Side milling** – vertical side face · **Straddle milling** – two parallel sides at once (2 side cutters)
- **Gang milling** – several cutters on one arbor, complex profile in one pass
- **Angular milling** – V-grooves, dovetail · **Form milling** – irregular contour, gears

6. Accessories & attachments

- **Arbor** – stub/long; Morse or ISO taper shank; key, spacers, bearing bush, draw-bar
- **Vices** – plain, swivel, universal (tilt) – hold work on table
- **Dividing (indexing) head** – divides periphery of work in equal parts; spiral-milling; plain, universal, optical
- **Rotary table** – circular slots, arcs, circular profile; worm & wheel drive (90:1)
- **Vertical milling head** – converts horizontal → vertical; swivels at angle
- **Slotting attachment** – converts rotary → reciprocating; keyways, splines, internal slots

7. Indexing (dividing head)

Indexing = dividing work periphery into **N equal parts** (gear teeth, splines, bolt heads, flutes) by turning work through equal angles 360°/N.

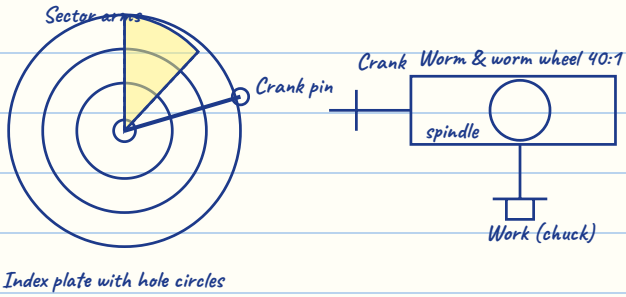


Fig 6: Dividing head – index plate & sector arms

(a) Direct indexing

- Index plate fixed on spindle (24 holes); plunger pin; worm disengaged
- Quick; only divisions = factors of 24 (2, 3, 4, 6, 8, 12, 24)

(b) Simple (plain) indexing

- Worm & wheel ratio **40:1** → 40 turns of crank = 1 rev of work
- Crank turns for N divisions:

$$n = 40 / N \text{ turns of crank}$$

- Index plates (B&S): P1: 15,16,17,18,19,20 · P2: 21,23,27,29,31,33 · P3: 37,39,41,43,47,49
- Method: fraction → full turns + (holes ÷ circle); set **sector arms** to count holes

Ex 1: N = 7 → $40/7 = 5\,{}^{5}/_{7} = 5 + {}^{15}/_{21}$ → **5 full turns + 15 holes in 21-hole circle**

Ex 2: N = 9 → $40/9 = 4\,{}^{4}/_{9} = 4 + {}^{12}/_{27}$ → **4 turns + 12 holes in 27-circle**

Ex 3: N = 28 → $40/28 = 10/7 = 1\,{}^{3}/_{7} = 1 + {}^{9}/_{21}$ → **1 turn + 9 holes in 21-circle**

Ex 4: N = 24 → $40/24 = 1\,{}^{2}/_{3} = 1 + {}^{10}/_{15}$ → 1 turn + 10 holes in 15-circle

(c) Compound indexing

- Two movements: crank rotated in plate (as simple) + **plate itself rotated** by 2nd pin (plate unlocked)
- $40/N = (a/b) \pm (c/d)$ → gets divisions not possible by simple e.g. 77, 91, 99

(d) Differential indexing

- Index plate is **geared to spindle** (gear train) – plate rotates automatically while crank turns
- Choose approx. number A (easy by simple indexing, near N):
- Crank turns = 40/A ; **Gear ratio = $40(A - N)/A$ = Driver/Driven**
- A > N → plate rotates same as crank; A < N → opposite (idler added)

(e) Angular indexing

- Dividing work by angle θ°: 1 turn of crank = 360/40 = **9°**

$$n = \theta^\circ / 9 \text{ (also 1 hole in 18-circle = } 0.5^\circ)$$

Ex: index 45° → 45/9 = 5 full turns; for 17°30′ = 17.5/9 = $1\,{}^{17}/_{18}$ → 1 turn + 17 holes in 18-circle (using 1 turn=9°; 1 hole in 18-circle = 0.5°)

Milling time & formulae

$$V = \pi DN/1000 \; ; \; f_m = f_t \times z \times N \text{ (mm/min)}$$

$$T = (L + A + O) / f_m \text{ min}$$

- L = length of work; A = approach; O = overrun (≈ 2–5 mm)
- Slab milling: $A = \sqrt{[d(D - d)]}$ (d = depth of cut, D = cutter dia)

- Face milling: $A = O = D/2$ (central) or $\approx \frac{1}{2}[D - \sqrt{(D^2 - W^2)}]$ (W = width)
- **MRR = $w \times d \times f_m$** (mm^3/min); N = rpm, z = teeth

> ★ *Important*
> - Simple indexing: $n = 40/N$ – learn 7, 9, 28 examples
> - Up milling vs down milling table – 5 marks every year
> - Time = $(L + A + O)/(f_t z N)$

GRINDING

8. Grinding machine – theory

Grinding = material removal by **abrasive grains** bonded in a wheel rotating at high speed (30–45 m/s); **very small chips**, fine finish (Ra 0.2–1.6 µm), tolerance ±0.002 mm; machines hard/hardened materials.

- Each abrasive grain = **negative-rake** (−30° to −60°) random cutting edge
- Chip formation: **rubbing → ploughing → cutting** (grain first rubs, then ploughs plastically, then cuts a chip)
- **Wheel wear**: (i) attritious wear – grain flattening (ii) grain fracture (iii) bond fracture / grain pull-out
- **Self-sharpening**: dull grain → force ↑ → grain fractures (new edges) or breaks out of bond → fresh grain exposed. Needs right **grade**

Classification of grinding machines

- **Precision**: cylindrical (external), internal, surface, centreless, tool & cutter, special (thread, gear, form, crankshaft, cam)

9. Cylindrical grinding machine

- Work between centres/chuck, rotates; wheel rotates (opposite at contact) ; table reciprocates
- **Plain** – wheel head fixed, table reciprocates (or traverse) ; **Universal** – table & wheel head swivel, internal grinding attachment, tapers
- Operations: **traverse** (long. feed), **plunge** (in-feed, no traverse – shoulders, forms), taper grinding

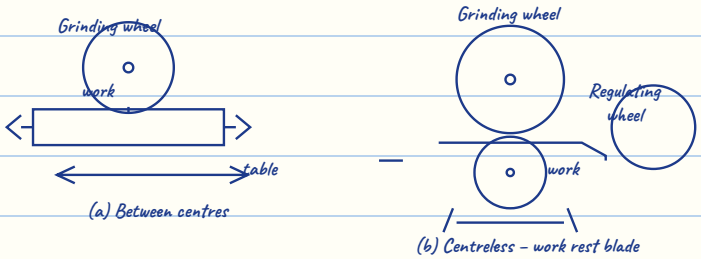


Fig 7: Cylindrical (centres) & centreless grinding

Centreless grinding

- No centres/chuck; work supported on **work-rest blade** between **grinding wheel** (fast) & **regulating wheel** (rubber-bonded, slow 20–100 rpm, tilted 1–5°)
- Regulating wheel controls work rotation & axial feed: $F = \pi D N \sin \alpha$
- + : no clamping, thin/long bars, high production, automatic, no deflection
- – : not for keyways / interrupted surfaces; setup costly

10. Surface grinding machine

- Produces flat surfaces; work on **magnetic chuck**/vice
- 4 types by wheel axis & table:

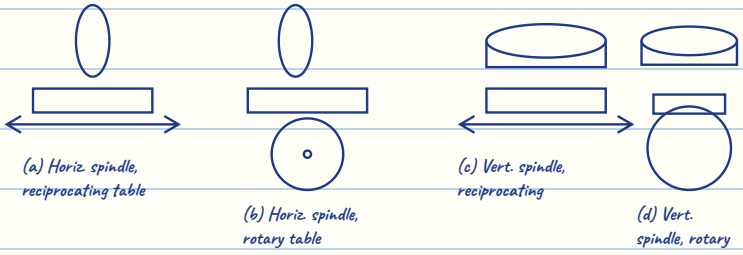


Fig 8: Four types of surface grinders

- (a) Horizontal spindle + reciprocating table – periphery of wheel; common, good finish
- (b) Horizontal spindle + rotary table – cylindrical/ring surfaces
- (c) Vertical spindle + reciprocating – cup/segmental wheel, large area, fast MRR
- (d) Vertical spindle + rotary table – production grinding of rings, washers

Internal grinding & Tool and cutter grinder

- **Internal**: small wheel in bore; chucking (work rotates), planetary (work fixed, wheel orbits), centreless internal
- **Tool & cutter grinder**: sharpens milling cutters, reamers, taps, drills, single-point tools; table swivels, work head, tailstock, universal wheel head; **cup / dish / saucer** wheels; tooth rest finger holds tooth at height of centre

11. Abrasives

Abrasive	Features / use
Aluminium oxide Al_2O_3	Artificial; tough; steels, steel alloys, high tensile; brown, white (friable), pink, ruby. Symbol A
Silicon carbide SiC	Harder but brittle; **cast iron, brass, Al, ceramics, carbide, non-ferrous**; green / black. Symbol C
CBN (cubic boron nitride)	Hardness ≈ diamond; HSS, hardened steels, Ni-superalloys; heat-resistant; costly
Diamond	Hardest; carbide, ceramics, glass, stone; not for steel (carbon affinity)
Others	Natural: emery, corundum, quartz, garnet; Zirconia alumina

➤ Mnemonic: **Steel → A**, **Cast iron/Carbide → C**

12. Bonds

Bond (symbol)	Features / uses
Vitrified (V)	Clay/glass fused; porous, strong, rigid, not affected by water/oil; 75 % wheels; max 35 m/s. Fragile (shock)
Silicate (S)	Sodium silicate; mild, grain released easily; cool cutting – edge tools, large wheels
Shellac (E)	Elastic; good finish; cam, crankshaft, roll grinding, thin wheels
Rubber (R)	Vulcanized rubber; elastic, thin cut-off wheels, regulating wheel of centreless
Resinoid (B)	Bakelite/synthetic resin; strong, up to 45–80 m/s; snagging, cut-off, rough grinding
Metal (M)	Bronze/nickel; for diamond & CBN wheels; long life

13. Grinding wheel specification

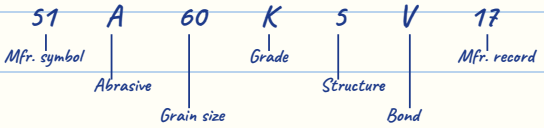


Fig 9: Marking system A 60 K 5 V (IS 551 / ISO)

➤ **A** = abrasive: A (Al_2O_3), C (SiC)

➤ **60** = grain size (mesh/in): coarse 10–24, medium 30–60, fine 70–180, very fine 220–600

➤ **K** = grade (bond strength): A–H soft, I–P medium, Q–Z hard

➤ **5** = structure (spacing): 1–8 dense (close), 9–15 open (wide)

- **V** = vitrified bond
- Read as: Al_2O_3 grain, medium-fine, medium grade, medium-dense, vitrified

★ *Important*

- Hard material → **soft grade, fine grain**; Soft/ductile material → hard grade, coarse grain, open structure
- Large contact area → soft grade, open structure; fine finish → fine grain

Wheel shapes (IS 2370)

- Straight (1), cylinder, tapered, straight cup, flaring cup, dish, saucer, segments, mounted points

Glazing & loading

- **Glazing**: grains dull, shiny surface; too hard grade → high temp, burn → dress
- **Loading**: pores clogged with metal chips (soft/ductile metals: Al, Cu); remedy – coarser/open wheel, coolant, dressing

14. Truing, dressing & balancing

Truing	Dressing
Restores **geometrical shape** & concentricity of wheel	Restores **cutting ability** – removes glazed grains / loaded chips
Done for new wheel / formed profile	Exposes fresh sharp grains
Tool: single-point **diamond**, diamond roll, form crushing roll	Tool: **star (Huntington) dresser**, abrasive stick, dressing stone, diamond
Removes more material	Removes only little layer

- **Balancing**: static balancing on balancing stand with ring-test; counter weights in flange slot → avoids vibration, chatter, wheel burst
- **Ring test** (tap – clear ring = no crack); use **blotters** & flanges (≥ ⅓ dia); wheel guard

FINISHING PROCESSES

15. Lapping

Lapping = abrasive finishing; loose abrasive **(Al_2O_3, SiC, diamond dust in oil/grease paste)** between work & a soft **lap** (cast iron, copper, lead), relative rubbing motion → removes 0.003–0.03 mm, tolerance ±0.0003 mm, Ra 0.025–0.2 μm.

16. Honing

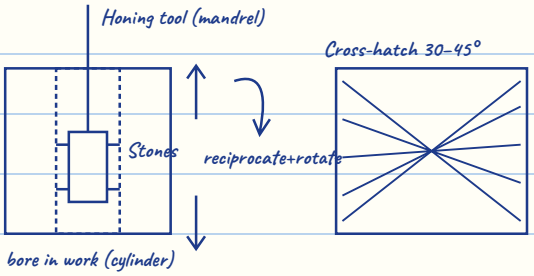


Fig 10: Honing – rotating + reciprocating hone & cross-hatch pattern

- **Honing** = low-speed finishing of **cylindrical bores** with bonded abrasive stones (hone) mounted in an expanding head; rotation (≈ 100–150 rpm) + reciprocation → **cross-hatch** 30–60° (retains oil)
- Removes 0.005–0.05 mm; Ra 0.1–0.4 µm; corrects **roundness, taper, straightness**
- Uses: engine cylinders, gun barrels, hydraulic cylinders, gears (hone)
- **Superfinishing (microfinishing)**: fine stone, light pressure (0.1–0.3 MPa), short **oscillating stroke** (1–5 mm) at 10–50 Hz + slow work rotation; lubricant (kerosene) → very smooth, Ra 0.01–0.1 µm; removes smeared layer; crankshaft, bearing races

17. Broaching

Broaching = a multi-tooth tool (**broach**) pushed/pulled past/through work in a **single stroke**; each tooth is higher than previous (**rise per tooth** 0.02–0.1 mm) → progressive cutting, finishes work in one pass.

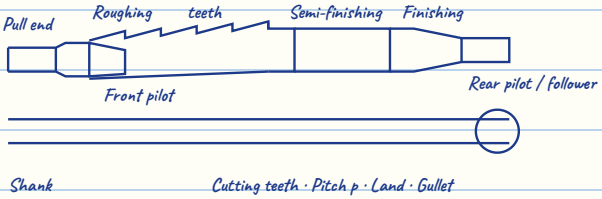


Fig 11: Parts of a broach

- **Parts**: pull end/shank, neck, front pilot, **roughing teeth**, **semi-finishing**, **finishing teeth**, rear pilot, follower rest/retriever
- **Nomenclature**: pitch ($p \approx 1.76\sqrt{L}$, L = cut length), land, gullet (chip space), face angle (rake 8–15°), back-off/clearance 1–3°, rise per tooth
- **Push type** – short, less than 8× dia (buckling) ; **Pull type** – tension, long, more accurate

Types of broaching

- **Internal**: round holes, keyways, splines, polygons (pull broach through pre-drilled hole)
- **Surface (external)**: flat, slots, contour, gun-slides; pull-up/pull-down, push; **continuous** (chain, rotary)
- **+ Advantages**: high production, high accuracy (±0.01 mm), good finish, roughing+finishing in 1 stroke, unskilled operator
- **– Limitations**: costly special tool, only through holes/surfaces, large force, uneconomical for small batches, no blind holes

18. Comparison: Grinding vs Lapping vs Honing

Point	Grinding	Lapping	Honing
Abrasive	Bonded wheel	**Loose** in paste on lap	Bonded stones
Speed	High (30 m/s)	Low	Low-medium
Pressure	High	Very light	Moderate
Metal removal	0.05–0.5 mm	0.003–0.03 mm	0.005–0.05 mm
Finish Ra (µm)	0.2–1.6	0.025–0.2	0.1–0.8
Surfaces	All (flat, cyl.)	Flat, cyl, spherical	Only bores (mostly)
Heat/stress	High, burn risk	Negligible	Low
Corrects	Size, finish	Flatness, finish	Roundness, taper, finish

JIGS & FIXTURES

19. Introduction & purposes

Jig = work-holding device that **locates, holds AND guides the tool** (bushes) – drilling, reaming, tapping. **Fixture** = locates & holds work, tool is **set** by feelers/set blocks, but **not guided** – milling, turning, grinding, welding.

Jig	Fixture
Guides tool (drill bush)	Only positions; tool set by gauge
Light, not clamped to table	Heavy, bolted to table (T-slot key)
Drilling, reaming, tapping	Milling, turning, grinding, broaching, welding

- **Purposes / uses**: eliminate marking-out & setting, **interchangeability**, accuracy, ↑ production, ↓ cost & time, unskilled labour, safety, less inspection, enable automation

20. Principles of location – 3-2-1 rule

- Free body has **12 degrees of freedom** (6 translational ±X,±Y,±Z + 6 rotational, i.e. 6 axial + 6 radial); clamping restrains the rest.
- **3-2-1 principle**: 3 pins on **primary** plane (restrain 3: 1 translation + 2 rotation) ; 2 pins on **secondary** plane (1 translation + 1 rotation) ; 1 pin on **tertiary** plane (1 translation) = **6 DOF**

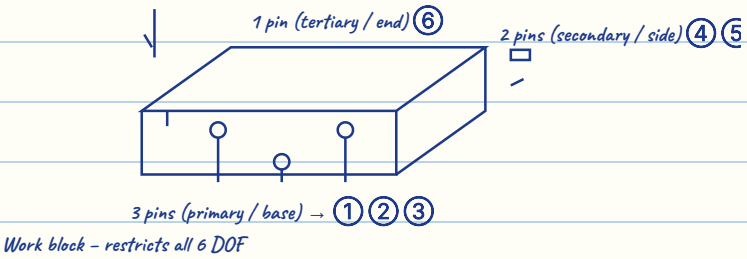


Fig 12: 3-2-1 principle of location (schematic)

Primary plane: **widest** area, 3 pins as far apart as possible → stability.
Secondary: 2 pins (long side). Tertiary: 1 pin (end).

Locating elements

- **Pins**: fixed, **button** (flat/round head), conical, **diamond pin** (relieved cylindrical, for hole positioning with 1 round + 1 diamond pin)
- **V-blocks**: centralise round work (self-centring, 90° / 120°), long/short
- **Rest buttons, pads, plates**, adjustable **jack pins** (rough castings), **nest**/cup locators
- **Locating from hole**: 1 cylindrical pin + 1 diamond pin; surface: ground flats

Principles of location & clamping

★ Important

- Locators on **machined** faces (datum), few, **sharp edges avoid**, chip-clearance, easily cleaned
- Clamp **against** locators – force directed on **supports**, not on unsupported work
- Clamp must be **quick, simple, strong**, not damage or distort work, not interfere with tool path
- Cutting force should be taken by locators/ fixed jaw – not by clamp
- Clamp: adequate force, **foolproof** (poka-yoke), loading/unloading easy

21. Clamping devices

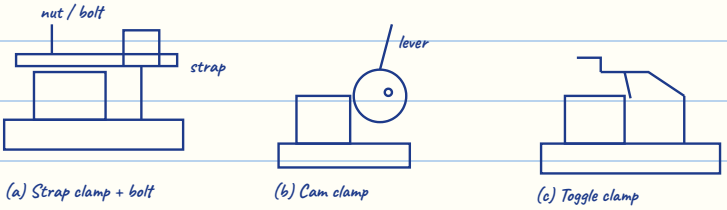


Fig 13: Strap, cam and toggle clamps

- **Strap (plate) clamp** – lever types (1st/2nd/3rd class), bolt+nut, simple, strong; slow
- **Screw clamp** – direct screw, swing/hook bolt, C-clamp, thumb screw; slow, large force
- **Cam clamp** – eccentric/spiral cam; **quick**, self-locking (angle ≤ friction), vibration may loosen; small stroke
- **Toggle clamp** – link mechanism, over-centre lock, quick, high force ratio; production/welding
- **Wedge clamp** – wedge/taper; self-locking if angle ≤ 2φ (≈ 6–8°) ; compact

22. Types of jigs

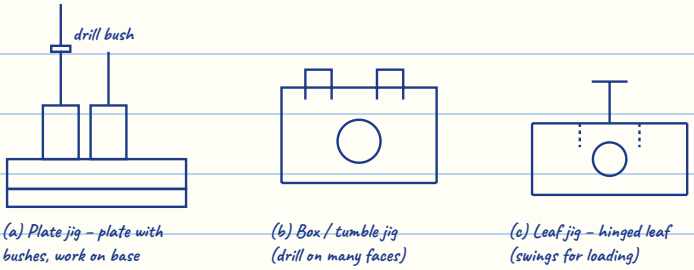


Fig 14: Plate, box and leaf jigs

- **Template jig** – simplest; plate/template laid on work, guides drill for hole pattern; not clamped; low accuracy, small batch, sheet metal
- **Plate jig** – plate with bushes + clamp; work clamped under; cheap, accurate
- **Box (tumble) jig** – box-shaped; work enclosed; rotated (tumbled) → holes on all sides without repositioning; chips need clearance
- **Channel jig** – channel section, work held by cam/screw, 1–2 sides
- **Indexing jig** – equally spaced holes on a circle; index plate + plunger
- **Diameter jig** – radial holes in round work (V-block); also trunnion/pot, sandwich jigs
- **Leaf jig** – hinged leaf (swing plate) for easy loading; small work; vibration–less rigidity

23. Types of fixtures

- **Milling fixture** – heavy; key way/tenon in T-slot; set block + feeler (tool setting); vice-type, indexing, string/gang; **straddle** milling
- **Turning (lathe) fixture** – on faceplate/chuck, balanced with counterweight; eccentric work, cranks, connecting rod
- **Grinding fixture** – magnetic/vacuum, rigid, indexing type
- **Welding fixture** – holds parts, resists distortion, toggle clamps, heat sink
- **Broaching fixture** – holds work in alignment with broach; spherical seat → self-aligning; take thrust

Bushings: press-fit (fixed), renewable, slip (liner), screw; hardened steel, clearance bush–tool 0.01–0.03; bush height ≈ 1–2 d above work.

24. Design principles checklist

★ *Important*

1. Study part drawing, operation sequence, batch size, m/c & tool, accuracy needed
2. Fix **datum** / locating surfaces (3-2-1); choose locators
3. Select **clamps** – force, direction, quick action
4. Ensure **rigidity** & strength; resist cutting force & vibration
5. **Foolproofing**: work fits only in one way; extra pins/ clearances
6. Provide **chip clearance**, easy cleaning, coolant drains
7. Easy loading/unloading, ergonomic, safe, light weight (jigs), balanced
8. Cost minimal; use standard parts; **material**: mild steel/cast iron, case-hardened locators (HRC 58–62), tool steel bushes
9. Tool guiding / setting block present; jig **feet** to rest on table

- **Advantages**: ↑ accuracy & interchangeability, ↓ time & cost, ↓ skill, ↑ safety, multi-machine operation
- **Limitations**: initial cost, design time, only economic for batch/mass production

25. Quick revision & likely exam questions

- Milling: $V = \pi DN/1000 \cdot f_m = f_z N \cdot T = (L+A+O)/f_m$
- Indexing: simple $n = 40/N$; angular $n = \theta/9$; differential ratio $= 40(A-N)/A$
- Wheel: **A 60 K 5 V** = abrasive, grain, grade, structure, bond
- Honing → bore, cross-hatch ; Lapping → loose abrasive, flat ; Broaching → one stroke, rise/tooth
- 3-2-1 = 6 DOF · Jig guides, Fixture doesn't

Probable questions

1. With neat sketch explain column & knee type milling m/c; list its parts (10M)
2. Classify milling machines. Differentiate up and down milling (10M)
3. Sketch & explain any 6 milling cutters; cutter nomenclature
4. What is indexing? Explain simple, compound, differential indexing; Find index movement for 7, 9, 28 divisions (10M)
5. Milling operations & accessories (dividing head, rotary table, vertical head)
6. Theory of grinding; self-sharpening; wheel wear
7. Explain cylindrical and centreless grinding with sketch; surface grinder types
8. Specify wheel A 60 K 5 V; abrasives & bonds; truing vs dressing; glazing & loading
9. Lapping, honing, superfinishing; compare grinding, lapping, honing
10. Broaching: types, broach nomenclature, advantages (10M)
11. Differentiate jig & fixture; explain 3-2-1 principle with sketch
12. Clamping devices; types of jigs and fixtures; design principles